

Seeking Principled Diffeomorphic Hylogisms (1/6): Constructing Smooth Quotients from the Mass-Configuration Space

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Errors found in this version (2026-05-29, Figshare v10). While the research note is intentionally speculative, several notational and technical gaps require attention. For example, the use of the Borel σ -algebra is yet unjustified (especially given the non-dense and pathological cases), there is a misapplication of “configuration space” to function spaces, a missing properness proof to support Ehresman’s theorem’s application, and the use of “Functorial” without the clear categorical context. None of these errors are necessarily terminal for the paper’s eventual goal but this note may change markedly in forthcoming iterations as errors are addressed.

Abstract

This research note is the first in a series that examines the bridge between the physical universe’s symmetric informational descriptions. En-route to a diffeomorphic connection from the state manifold to the semantic vector space, this note seeks a functorial hylogism, allowing the principled identification of physical structures across space and scales. As an observation window’s content, bounds, or resolution deforms, the identity of viewed structures changes smoothly. The note discusses the pre-requisites for such smooth identity changes.

1 Introduction

1.1 Internal Terminology

The so called “it-from-bit” debate contests whether information is ontologically prior to reality. Whereas informational realists argue reality is information-theoretic in origin, physicalists maintain that matter instantiates information [1], [2], [3]. This note does not try to settle that debate. Instead, this note investigates the feasibility of a diffeomorphic¹ correspondence between each universe, under the (progenitor-agnostic) assumption that the pair are at least analytically and discursively distinct. Denote the pair:

$$P\text{- the physical universe, } I\text{- the informational universe.} \quad (1)$$

An *hylogism*² maps between a physical and informational element. To imagine, measure, or articulate a physical system is to instantiate information in the physical universe. One may also move from an information-theoretic description of a physical scene to a synonymous non-physical descriptor. Rather than the exact physical description (mass distribution, coordinate system, velocity fields, etc.), this note seeks an informational element that facilitates efficient calculation and identification.

¹“Diffeomorphic” is a scary way of saying “smooth and without tearing”. So, a smooth, continuous deformation of matter will cause a smooth, continuous deformation of the informational encoding. That quality is not common in current computer vision classifiers.

²Also “hylogical correspondence”. “Hylogics” is the combination of the ancient Greek “hyle”, referring to matter, and “logos”, referring to words or reason. Here, the author uses “hylogic” as a 3-syllable shorthand meaning literally “physical-informational”.

To clarify, introduce the following pipeline and consistent shorthand terminology:

$$H : \underbrace{P \longleftrightarrow P \rhd I}_{\text{Semiotic Hylogics } (H_S)} \overset{\text{Transductive Hylogics } (H_T)}{\longleftrightarrow} \underbrace{I \rhd P \longleftrightarrow I}_{\text{Functorial Hylogics } (H_F)} \quad (2)$$

Definition 1 (Hylogics). *The overarching study of correspondences $h \in H$ between the physical universe (P) and the informational universe (I).*

The pipeline is:

1. a physical scene and its observer exist in P ,
2. the observer conceptualises/measures the physical scene $P \rhd I$,
3. the physical scene takes an information-theoretic form $I \rhd P$,
4. the information-theoretic form morphs to symmetric objects in I .

Term the study of the former correspondences $h_s \in H_s$ *semiotic hylogics*, the boundary translation $h_t \in H_T$ *transductive hylogics*, and the continuous semantic mappings $h_f \in H_F$ *functorial hylogics*.

Definition 2 (Semiotic Hylogics). *The study of how the physical universe instantiates information from raw physical states to physically encoded signs ($P \longleftrightarrow P \rhd I$). Semiosis studies how information is locally enacted by a physical observer [4], [5], [6].*

Definition 3 (Transductive Hylogics). *The study of the boundary and translation mechanisms that convert physically-instantiated information into an information-theoretic structure ($P \rhd I \longleftrightarrow I$). The correspondence $h_t \in H_T$ is, here, a special case that bridges the physical act of measurement and the informational encoding of a physical system ($P \rhd I \longleftrightarrow I \rhd P$).*

Definition 4 (Functorial Hylogics). *The symmetry-preserving translations of an information theoretic physical state within the informational universe³ ($I \rhd P \longleftrightarrow I$)*

Distinguishing these ideas with different definitions is useful in categorising an otherwise fragmented literature [7], [8], [9], [10]. This paper seeks a smooth, functorial hylogism h_F from a physical information-encoding to other non-physical descriptions of that scene. Functorial hylogisms are useful for, say, examining the symmetry groups of semantic concepts. For example, many physical systems are assigned the string “chair”, and one might seek a principled method for determining when a physical form stops being a chair under deformation. Alternatively, one may seek a reverse mapping from “chair” to the pre-image of physical “chair-states”.

1.2 H_F Components: Field Interpretation

1.2.1 Mass-Configuration Space $\Gamma(P, \mathbb{S}_0)$

Formally, if P is a D -dim. physical space with a coordinate system and volume element dV , a mass configuration on P is a mass-density scalar field:

$$\rho : P \rightarrow \mathbb{R}_{\geq 0} \quad (3)$$

where $\rho(x)$ gives the mass density at position $x \in P$ (e.g., $x = [x_1, \dots, x_D]$). The configuration space of admissible mass-density distributions is thus (append. A):

³e.g., between a complete geometric description and a coordinate-free semantic representation.

$$\Gamma(P, \mathbb{S}_0) = \left\{ \rho : P \rightarrow \mathbb{R}_{\geq 0} \mid \int_P \rho(\mathbf{x}) dV = M, \rho \text{ satisfies the relevant physical constraints} \right\}. \quad (4)$$

where each point is a complete mass distribution over P . Infinitesimal directions at a configuration ρ are perturbations $\delta\rho$ of the density field. If total mass is fixed, these perturbations satisfy⁴:

$$\int_P \delta\rho(\mathbf{x}) dV = 0. \quad (5)$$

While these constraints support physical feasibility, later sections relax configuration constraints to ensure the flatness and trivialisability of their resultant quotients. This note gives a pre-construction⁵ for a matter-only diffeomorphic hylogism $h_{0 \rightarrow \mathbb{V}}$ between $\Gamma(P, \mathbb{S}_0)$ and a symmetry-preserving description elsewhere in the informational universe I . Specifically, the semantic vector space $\mathbb{V} \subset I$ ⁶:

$$h_{0 \rightarrow \mathbb{V}} : \Gamma(P, \mathbb{S}_0) \leftrightarrow V \quad (6)$$

The mass distribution and coordinate system are not a complete description of the physical universe, but distinguishing $\Gamma(P, \mathbb{S}_0)$ allows for the identification of static physical objects, without regard to non-mass features. For example, a chair might have some velocity – its particles some spin – not accounted for by mass alone, but the chair’s identity is (arguably) an invariant over these properties.

1.2.2 General Configuration Spaces

To generalise in the current early context, let these non-mass components distinguish two otherwise identical mass distributions within a finer, more pluralistic, physical state-configuration space $\Gamma(P, \mathbb{S})$. The section mapping becomes $\mathbb{S}_{\rho, \Phi}$ (hereafter $\mathbb{S}$):

$$(\rho(\mathbf{x}), \Phi^{(1)}(\mathbf{x})) \neq (\rho(\mathbf{x}), \Phi^{(2)}(\mathbf{x})) \implies s^{(1)} \neq s^{(2)} \quad \text{for } s^{(1)}, s^{(2)} \in \mathbb{S} \quad (7)$$

Now, assume the physical state space may distinguish between any two physical scenes⁷. The former mass-configuration space is the base of a fibre bundle feeding up to $\Gamma(P, \mathbb{S})$ (figure 1). Notice that this continuous, fine-grained state space cannot discretely partition to objects in I while h is diffeomorphic; a smooth state change δs is discordant to an abrupt information change Δi , $i \in I$. So, partition the information space into a continuous and discrete representation. This note uses the continuous semantic vector space $\mathbb{V}$ and the discrete SVS lattice $\mathcal{L}(\mathbb{V})$ ⁸.

⁴When the path represents physically continuous redistribution rather than an abstract change of description, the density evolution may be constrained by a velocity field v_t through

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0,$$

(conservation of mass along a trajectory $\gamma(t) = \rho_t$.)

⁵This note does not yet fully prove the existence of $h_{0 \rightarrow \mathbb{V}}$.

⁶Subset/Subclass notation applies loosely here to mean that the semantic vector space is somehow contained in I .

⁷A false assumption used for simplicity.

⁸A “lattice” in discrete group theory is a set of points that mask an otherwise continuous or compact group, think of the integers $\mathbb{Z}$ over the reals $\mathbb{R}$ [11]

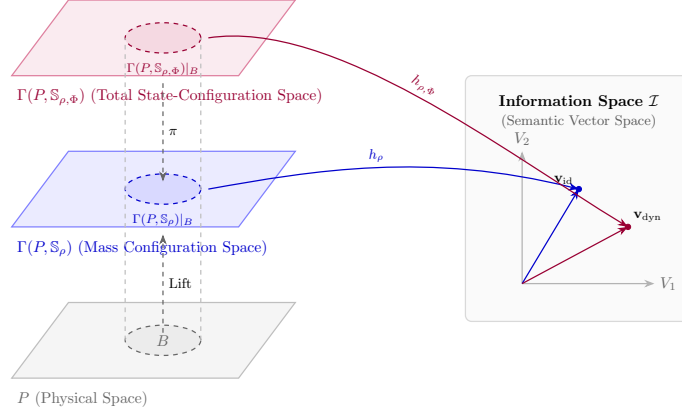


Figure 1: Hylogic correspondences: the physical universe P permits (coordinate-dependent) mass-density permutations in $\Gamma(P, \mathbb{S}_0)$, further refined by non-mass components in the state configuration space $\Gamma(P, \mathbb{S})$.

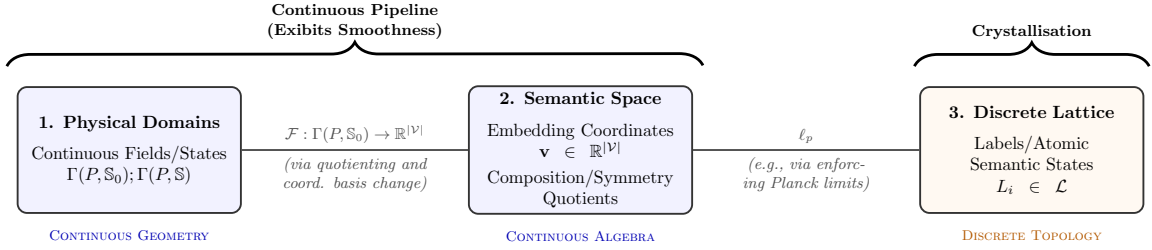


Figure 2: 3-Step Transition from continuous physical field and semantic embeddings, to discrete logical identifiers.

1.2.3 Semantics and Discrete Lattice Formation

Typically, one defines the lattice empirically by, say, grouping “chair-like” semantic vectors to the term “chair”. There must normally exist, therefore, an external vocabulary of discrete labels that the SVS trains towards to preserve desirable syntactic or semantic relations in the continuous word-encodings [12], [13], [14]. This training usually depends on the frequency or collocation of tokens in a corpus [15], [16], [17].

In the current context then, one would fix the physical arrangements in $\Gamma(P, \mathbb{S}_0)$ or $\Gamma(P, \mathbb{S})$ along with the labels in $\mathcal{L}$. Then, one would optimise the physical-vector and vector-discrete correspondences according to the experimentalist’s preferences. This practice is ubiquitous in modern machine learning and underpins vector embeddings in language models [18], [19].

Yet, notice that objects in nature do not have labels. Humans go forth making different observations and conjuring different perceptions of an object, each internal cerebration itself a physical signature. These mental labels may then be written or otherwise physically instantiated, but neither the written nor the mental label need be similar to the described object. “Why then” one asks “do we cling to those arbitrary names?”. “Why don’t we call objects by, say, their exact cubic volume – pass me that 125cm^3 or that 7.0432 cm^3 ? Why don’t we place a thermometer at the furthest point within the object’s surface and call a thing 17°C ?” These names, by dint of their physical descriptiveness, offer a more principled convention than our (often arbitrary) familiar labels.

Like an alien species, humans might come to talk of objects by their physical identifiers, with the label for “chair” and “table” more syntactically similar than those for, say “chair” and “hair”. Humans might also divine a method for measuring the inaccuracy of language relative to that principled system. Suppose, for example, that halving an object would immediately halve the object’s name so that the two half-names could serve as unique standalone identifiers. Suppose recombining the object could recombine the names⁹. In this mirrored object-identifier duality,

⁹(or, vice versa, the recombined names would immediately refer to the whole object).

a systematic correspondence exists between spatial structures and their symbolic identifiers. This systematisation is important, not just for generating consistent labelling conventions, but also for tracking structures computationally as they metamorphose.

2 Current Literature

That a physical object’s structure may entail its principled, mathematical name is an old idea. Atiyah and Singer (1968) proved that continuous differential operators relate to discrete integer invariants [20]. More recently, in Topological Deep Learning (TDL), researchers have mapped topological object-data to identifiers like Betti numbers [21]. Since a physical scene is identifiable via more than topology (or even geometry in the GDL mapping case [9]), a full physical-informational mapping may yield a more comprehensive naming method.

Earlier works do not typically instantiate the full hylogical pipeline because each field captures a local projection of the problem. Computer vision and open-vocabulary 3D mapping connect sensed scenes or reconstructed fields to semantic features [22], [23], [24], [25], [26]. GDL formalises symmetry-aware representation [9], while TDL formalises learning over topological domains [21]. Shape analysis formalises deformation and identity [27]; density geometry formalises physical configuration spaces through diffeomorphism groups and spaces of densities [28]; shape calculus studies smoothly moving observation domains [29]; and bisimulation, quotient learning, and quotient generative modeling formalise when detailed states may collapse without losing relevant structure [30], [31], [32], [33]. Symbol-grounding theory motivates the world-to-meaning bridge [34]. What remains for the current purpose is to organise these fragments into one physical-informational pipeline.

The reader should interpret this note’s framing with care. Since many of this note’s technical ingredients already exist in neighbouring literatures¹⁰, the note may seem to overclaim. Yet, aggregating these methods within the overarching hylogical pipeline is genuinely useful. Physical mass configurations, observer-relative distinguishability, and semantic representations are unifiable, and the claim here is that the hylogical problem-framing allows disparate components to be compared and eventually connected. This synthesis allows the study of an overarching, locally smooth correspondence between physical structures and their description. Perhaps such efforts will pave the foundation for a computable ‘universal catalogue of nature’.

3 Formal Construction

This formal construction follows two threads: the first is a discussion of claims that may hold in the general case, and the second is a specific simplified example. So, the construction’s actual yield is a minimal viable proof-of-concept – conjectural claims peripheral.

3.1 Volumetric Distinguishability

Within the present mass-density representation, object identity is not primitive. A configuration $\rho \in \Gamma(P, \mathbb{S}_0)$ specifies how mass is distributed over P , but does not determine which sub-regions count as objects. To distinguish objects, consider drawing volumes of interest:

$$B \subset P, \quad B \in \mathcal{B}(P) \tag{8}$$

where $\mathcal{B}(P)$ is the Borel σ -algebra¹¹ of the underlying physical space P . Moving to the configuration space $\Gamma(P, \mathbb{S}_0)$, the Borel sets induce the restricted density field:

¹⁰(e.g., quotient spaces, shape spaces, density geometry, geometric deep learning, semantic embeddings, and symbol grounding)

¹¹A Borel σ -algebra is a mathematical collection of all the neatly measurable regions one can safely slice a space into [35]. The Borel sets may be handy later for integrating/gluing bubbles together.

$$\rho|_B : B \rightarrow \mathbb{R}_{\geq 0}, \quad (9)$$

whose graph is measurable¹². Liken this process to taking a variable convolution kernel across the whole of a D-dimensional reality, but rather than discrete operations encoding each volume, equivalence classes across unseen regions map smoothly via coordinate transforms.

3.1.1 Distinguishability on the Configuration Space

Here, the choice of formalism forks. On one hand, mass captured in the ‘Borel bubbles’ could be a point in a quotient space (fig. 3).

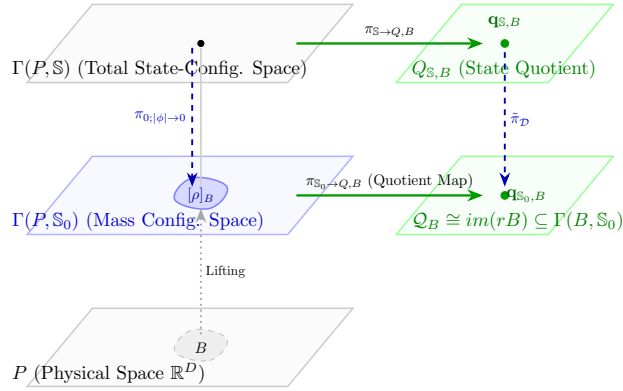


Figure 3: The original 3-part physical, configuration, and full-state distinction. $\mathcal{P}$ ’s volumes of interest B lift to an ‘observer-ignorance’ equivalence class on the configuration space $[\rho]_B$. That class compresses to a single quotient point $\pi_{S \rightarrow Q, B}([\rho]_B) \in Q_{S, B}$. Moving in $Q_{S, B}$ yields different mass configurations within a static B volume/coordinate set.

On the other hand, one could introduce a new mass-configuration space just for the identified B-regions, $\Gamma(B, S_0)$ (fig. 4). The combination of these local, induced regions is then a sheaf-like system.

¹²Simply, taking a cookie-cutter to a mass distribution produces ‘strings’ (relations) between points in $\mathcal{P}$ and the mass-distribution scalar field value ρ at those coordinates. The collection of those coordinate-density relations is a measurable restricted graph.

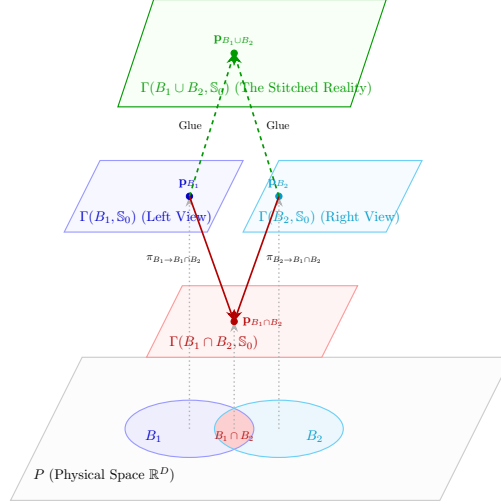


Figure 4: The individual volumes of interest, their intersection, and their union each have a matter-configuration space. Independent local mass-configurations must satisfy restriction mappings to their spatial intersections before they can be rigorously glued into a unified object space.

The independent local spaces $\Gamma(B_1, \mathbb{S}_0)$ and $\Gamma(B_2, \mathbb{S}_0)$ are evaluated at their spatial overlap. If the mass configurations reach a consensus within the intersection spaces $\Gamma(B_1 \cap B_2, \mathbb{S}_0)$ (verified via downward restriction maps), one glues the points upwards to form a stitched, multi-volume reality in $\Gamma(B_1 \cup B_2, \mathbb{S}_0)$.

Applying strict local sheaf restrictions to a dynamic physical state space introduces a boundary error: a local restriction mandates that the environment beyond the Borel set formally ceases to exist ($\rho : B \rightarrow \mathbb{R}_{\geq 0}$)¹³. Thus, if a continuous deformation pushes a physical mass across the boundary of B , the mass leaves the space, violating mass conservation¹⁴. To preserve the continuity of the global physical environment, this framework abandons the local sheaf in favor of the state quotient space $Q_{\mathbb{S}, B}$. Rather than truncating the universe at the boundary of B , the quotient map maintains the global mass configuration $\Gamma(P, \mathbb{S}_0)$ but mathematically collapses all mass variations occurring *outside* B into a single observer-ignorance equivalence class.

To see how, imagine that the Borel sets could ensnare any possible physical object across all scales. Once caught, an object's peripheral environment is assumed irrelevant to the object's identity¹⁵. So, the equivalence class of mass configurations that preserve the focal 'bubbled mass' is a collection of parallel universes where an object is still present and true. Compress this stack of worlds into a single point. Now 'wiggle' the Borel set's contents, perturbing the mass distribution. A new equivalence class of parallel universes preserves this new ensnared object. Compress again to a point. Continuing repeatedly thus yields a space of all such compressed points: the *Borel-induced mass quotient space*, $Q_{\mathbb{S}_0, B}$ (specific to the current Borel bubble(s)'s walls.)

Then, deforming the Borel container instantiates a new quotient space $Q_{B'}$. Repeatedly quotienting the mass-configuration space $\Gamma(P, \mathbb{S}_0)$ in this way gives $\mathcal{E}_\rho$, the *disjoint Borel-induced mass quotient union*, or *coproduct* over the Borel σ -algebra:

$$\mathcal{E}_\rho = \coprod_{B \in \mathcal{B}(P)} Q_{\mathbb{S}_0, B} \quad (10)$$

Term $\mathcal{E}_\rho$ the ' ρ -catalogue'. At first glance, this aggregation appears similar to a sheaf. However, the disjoint union

¹³The sheaf style is also problematic for facilitating smooth, differentiable changes to the Borel set and its contents. This will bedevil attempts at, say, backpropagating through the diffeomorphism to optimise structures.

¹⁴Perhaps useful in modelling parallel universes where mass is not conserved.

¹⁵A chair is a chair whether a researcher sits, stands, or, from afar, expounds philosophically on it!

does not topologically stitch or glue these spaces together to reconstruct the configuration space. Because every quotient space $Q_{\mathbb{S}_0, B}$ is a projection derived from the exact same, unbroken global mass configuration $\Gamma(P, \mathbb{S}_0)$; physical reality remains intact.

In fact, enforcing specific properties on the base space, mass configuration space, and Borel set deformations can ensure that parts of this disjoint union are smooth in both the horizontal and vertical directions¹⁶. To show this statement holds, assume a new smooth, continuous scalar field α (θ -parameterised) on the base space.

$$\alpha_\theta : P \rightarrow \mathbb{R} \quad (11)$$

Now cut that scalar field at a threshold altitude c :

$$B_{\theta, c} = \{\mathbf{x} \in P : \alpha_\theta(\mathbf{x}) \geq c\} \quad (12)$$

So, the P -region with α -values lying above the threshold c is a Borel-proxy. Any region $B_{\alpha, c}$ is automatically a Borel¹⁷. Happily, α can vary smoothly (though splitting events must be excluded from the regular deformations). Denote the smooth hypersurface:

$$\partial B_{\theta, c} = \alpha_\theta^{-1}(c) \quad (13)$$

Note that this smooth framing does not permit single point-wise admission into $B_{\theta, c}$, nor kinked boundaries. Yet, α_θ approximates these regional features with sufficiently fine parameterisations in the limit. For each c , define:

$$\rho_1 \sim_{\theta, c} \rho_2 \iff \rho_1|_{B_{\theta, c}} = \rho_2|_{B_{\theta, c}}. \quad (14)$$

Quotienting over all non-bubbled mass-configurations yields:

$$Q_{\theta, c} = \Gamma(P, \mathbb{S}_0) / \sim_{\theta, c}. \quad (15)$$

The disjoint union over regular cuts c is:

$$E_\rho = \coprod_{(\theta, c) \in \mathcal{A}_{\text{reg}}} Q_{\theta, c}, \quad (16)$$

where:

$$\mathcal{A}_{\text{reg}} = \{(\theta, c) : c \text{ is a regular value of } \alpha_\theta\}. \quad (17)$$

The regular c values of each α -field are those rendering the resulting bubbles smooth. Yet, at maxima or infinite level sets, non-regular bubbles occur. To obtain a singleton, consider the limiting case:

$$\lim_{c \uparrow a(\mathbf{x}_p^*)} \{\mathbf{x} : a(\mathbf{x}) \geq c\}, \quad (18)$$

¹⁶i.e., deforming mass in the bubble and deforming the bubble boundaries induces smooth moves in every direction of the disjoint union - the catalogue is less a book of discrete pages and more a connected space.

¹⁷The reverse is not necessarily true.

where $\mathbf{x}_p^*$ is a local maximum. Thus point-observation is either a limiting idealisation or a finite-resolution cap around the peak. At this juncture, one must still prove that smooth local (θ, c) movements cause analogous $Q_{\theta,c}$ moves. If one restricts the focus to bubbles that lie inside a common compact region, Ehresmann's lemma[36]¹⁸ places the restricted Borel set space atop a locally trivial fibration of the regular bubble-parameter space¹⁹:

$$\pi_B : \mathcal{B} \rightarrow \mathcal{A}_{reg} \quad (21)$$

$$\text{where } \pi_B^{-1}(\theta, c) = B_{\theta,c} \quad (22)$$

Local triviality says that a local bubble $B_{\theta,c}$ is a smooth deformation of a reference bubble B_{θ_0,c_0} :

$$\pi_B^{-1}(A) \cong A \times B_{\theta_0,c_0} \quad (23)$$

where $A \subset \mathcal{A}_{reg}$ is the differential neighbourhood of (θ_0, c_0) . So, for each nearby $(\theta, c) \in A$, there is a smoothly varying diffeomorphism

$$\psi_{\theta,c} : B_{\theta_0,c_0} \rightarrow B_{\theta,c}. \quad (24)$$

Thus, regular bubbles form a locally trivial smooth family over $\mathcal{A}_{reg}$; bubbles do not vary as unrelated Borel sets. Local bubble trivialisation induces further local trivialisation of the quotient fibres. Indeed, a quotient class

$$[\rho]_{\theta,c} \in Q_{\theta,c} \quad (25)$$

arises from the observed restriction $\rho|_{B_{\theta,c}}$ using the bubble diffeomorphism $\psi_{\theta,c}$. Represent this observed density on the fixed reference bubble by

$$\rho|_{B_{\theta,c}} \circ \psi_{\theta,c}. \quad (26)$$

Thus, nearby quotient fibres are smoothly identified with a common reference fibre. Locally,

$$E_{\rho}^{\text{reg}} \cong A \times Q_{\theta_0,c_0}. \quad (27)$$

Hence the regular observer quotient union is smooth in both directions: horizontal motion corresponds to smooth motion in the cut-parameters (θ, c) , while vertical motion corresponds to smooth variation of the observed mass-density class inside a fixed quotient fibre.

¹⁸More simply, a smooth, ‘point-to-point’, differentiable, ‘nicely invertible’ map from a total space to a base space is a ‘well-behaved’ fibration, ensuring the transfer of nice smooth properties from the bubble-parameter space $(\theta, c) \in \mathcal{A}_{reg}$ to the top space B of all regularly reachable bubble coordinates.

¹⁹Fixing (θ, c) yields all points surviving the α -cut.

$$\mathcal{B} = \{(\theta, c, \mathbf{x}) : \alpha_{\theta}(\mathbf{x}) \geq c\} \quad (19)$$

$$(\mathcal{B})_{\theta,c} = \{\mathbf{x} \in P : \alpha(\mathbf{x}) \geq c\} = B_{\theta,c} \quad (20)$$

References

- [1] J. A. Wheeler, “Information, physics, quantum: The search for links,” in *Complexity, Entropy, and the Physics of Information*, W. H. Zurek, Ed., Redwood City, CA: Addison-Wesley, 1990, pp. 3–28.
- [2] R. Landauer, “Information is physical,” *Physics Today*, vol. 44, no. 5, pp. 23–29, 1991.
- [3] C. Rovelli, “Relational quantum mechanics,” *International Journal of Theoretical Physics*, vol. 35, no. 8, pp. 1637–1678, 1996.
- [4] F. de Saussure, *Course in General Linguistics*, C. Bally and A. Sechehaye, Eds., trans. by W. Baskin. New York: Philosophical Library, 1959, Originally published in French as *Cours de linguistique générale* (1916).
- [5] C. S. Peirce, *Collected Papers of Charles Sanders Peirce* (Elements of Logic), C. Hartshorne and P. Weiss, Eds. Cambridge, MA: Harvard University Press, 1932, vol. 2.
- [6] H. H. Pattee, “The physics of symbols: Bridging the epistemic cut,” *Biosystems*, vol. 60, no. 1-3, pp. 5–21, 2001.
- [7] J. G. Roederer, *Information and its Role in Nature*. Berlin, Heidelberg: Springer, 2005, See specifically Chapter 1, where Roederer critiques the fragmented definitions of information across physics, biology, and computation.
- [8] L. Floridi, *The Philosophy of Information*. Oxford University Press, 2011, Floridi documents the conceptual fragmentation of information theory and explicitly calls for a unified structural framework.
- [9] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković, *Geometric deep learning: Grids, groups, graphs, geodesics, and gauges*, 2021. arXiv: [2104.13478](https://arxiv.org/abs/2104.13478) [cs.LG].
- [10] D. I. Spivak, *Category Theory for the Sciences*. MIT Press, 2014.
- [11] D. Micciancio, *CSE 206A: Lattice Algorithms and Applications*. University of California, San Diego (UCSD), 2012, Lecture 1: Introduction to Lattices. Accessed: May 21, 2026. [Online]. Available: <https://cseweb.ucsd.edu/classes/wi12/cse206A-a/lec1.pdf>.
- [12] Y. Bengio, R. Ducharme, P. Vincent, and C. Jauvin, “A neural probabilistic language model,” *Journal of Machine Learning Research*, vol. 3, pp. 1137–1155, 2003.
- [13] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean, “Distributed representations of words and phrases and their compositionality,” in *Advances in Neural Information Processing Systems*, vol. 26, 2013.
- [14] J. Pennington, R. Socher, and C. D. Manning, “GloVe: Global vectors for word representation,” in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, Doha, Qatar: Association for Computational Linguistics, 2014, pp. 1532–1543. DOI: [10.3115/v1/D14-1162](https://doi.org/10.3115/v1/D14-1162).
- [15] Z. S. Harris, “Distributional structure,” *Word*, vol. 10, no. 2–3, pp. 146–162, 1954. DOI: [10.1080/00437956.1954.11659520](https://doi.org/10.1080/00437956.1954.11659520).
- [16] J. R. Firth, “A synopsis of linguistic theory, 1930–1955,” in *Studies in Linguistic Analysis*, Oxford: Blackwell, 1957, pp. 1–32.
- [17] P. D. Turney and P. Pantel, “From frequency to meaning: Vector space models of semantics,” *Journal of Artificial Intelligence Research*, vol. 37, pp. 141–188, 2010. DOI: [10.1613/jair.2934](https://doi.org/10.1613/jair.2934).
- [18] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA: MIT Press, 2016.
- [19] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Association for Computational Linguistics, 2019, pp. 4171–4186. DOI: [10.18653/v1/N19-1423](https://doi.org/10.18653/v1/N19-1423).
- [20] M. F. Atiyah and I. M. Singer, “The index of elliptic operators: I,” *Annals of mathematics*, vol. 87, no. 3, pp. 484–530, 1968.
- [21] M. Hajij et al., *Topological deep learning: Going beyond graph data*, 2022. arXiv: [2206.00606](https://arxiv.org/abs/2206.00606) [cs.LG].
- [22] A. Radford et al., “Learning transferable visual models from natural language supervision,” in *Proceedings of the 38th International Conference on Machine Learning*, 2021, pp. 8748–8763.
- [23] N. M. M. Shafiullah, C. Paxton, L. Pinto, S. Chintala, and A. Szlam, “CLIP-Fields: Weakly supervised semantic fields for robotic memory,” in *Robotics: Science and Systems*, 2023.

- [24] J. Kerr, C. M. Kim, K. Goldberg, A. Kanazawa, and M. Tancik, “LERF: Language embedded radiance fields,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.
- [25] S. Peng, K. Genova, C. Jiang, A. Tagliasacchi, M. Pollefeys, and T. Funkhouser, “OpenScene: 3d scene understanding with open vocabularies,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 815–824.
- [26] K. M. Jatavallabhula et al., “ConceptFusion: Open-set multimodal 3d mapping,” in *Robotics: Science and Systems*, 2023.
- [27] N. Charon and L. Younes, *Shape spaces: From geometry to biological plausibility*, 2022. arXiv: [2205.01237 \[math.DG\]](#).
- [28] B. Khesin, G. Misiołek, and K. Modin, *Information geometry of diffeomorphism groups*, 2024. arXiv: [2411.03265 \[math.DG\]](#).
- [29] M. Berggren, “Shape calculus for fitted and unfitted discretizations: Domain transformations vs. boundary-face dilations,” *Communications in Optimization Theory*, pp. 1–33, 2023, Article ID 27.
- [30] G. J. Pappas, “Bisimilar linear systems,” *Automatica*, vol. 39, no. 12, pp. 2035–2047, 2003. DOI: [10.1016/S0005-1098\(03\)00255-3](#).
- [31] É. Mehr, A. Lieutier, F. S. Bermudez, V. Guitteny, N. Thomé, and M. Cord, “Manifold learning in quotient spaces,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 9165–9174.
- [32] T. Yokota and H. Hontani, *Manifold modeling in quotient space: Learning an invariant mapping with decodability of image patches*, 2022. arXiv: [2203.05134 \[cs.CV\]](#).
- [33] Y. Xu et al., *Quotient-space diffusion models*, 2026. arXiv: [2604.21809 \[cs.LG\]](#).
- [34] S. Harnad, “The symbol grounding problem,” *Physica D: Nonlinear Phenomena*, vol. 42, no. 1–3, pp. 335–346, 1990. DOI: [10.1016/0167-2789\(90\)90087-6](#).
- [35] C. Walkden, *MAGIC010 Ergodic Theory*. Mathematics Access Grid Instruction and Collaboration (MAGIC), 2013, Lecture 3: Invariant Measures. From the 10-credit MAGIC course on Ergodic Theory. Accessed: May 21, 2026. [Online]. Available: <https://personalpages.manchester.ac.uk/staff/charles.walkden/magic/lecture03.pdf>.
- [36] C. Ehresmann, “Les connexions infinitésimales dans un espace fibré différentiable,” 1950.

A Notation

A.1 Base Quotient Methods

A distinction exists between methods that (i) quotient the base and field fibres directly, and (ii) quotient over a pre-defined configuration equivalence class. Table 2 elicits that distinction.

²⁰This notation is vulnerable to unstructured component-dropout. A multi-index notation is more precise, but is too involved for current purposes.

Concept	Notation	Example / Context
Fibre Bundle	$\mathbb{S}_\infty$ or $\mathbb{S}$	The infinite-dimensional state space of field values at $\mathbf{x}$ ($\varprojlim \mathbb{S}_{[\underline{\phi}; \overline{\phi}]}$).
Sub-Bundle	$\mathbb{S}_{\underline{\phi}; \overline{\phi}}$	The finite approximation of state space including only components $[\underline{\phi}, \overline{\phi}]$. ²⁰
Configuration Space	$\Gamma(P, \mathbb{S}_{\underline{\phi}; \overline{\phi}})$	The space of all P-spanning fields in the section $\mathbb{S}_{\underline{\phi}; \overline{\phi}}$. One element is a P-spanning field or a collection of P-spanning fields.
Spatial Dimensions	S_{dim}	S_x, S_y (Distinct from state space $\mathbb{S}$).
Elements	$s_{[\underline{\phi}; \overline{\phi}]}$ or s	$s_{[\underline{\phi}; \overline{\phi}]} \in \mathbb{S}_{\underline{\phi}; \overline{\phi}}$ (Inherits filter subscript; dropped to s when context is unambiguous).
Element Components	s_{dim}	$s_{\mathbf{x}}$ or s_ρ (An internal coordinate/dimension of s , denoted by subscript).
Instances / Obs.	$s_F^{(obs)}$	$s_0^{(1)}, s_0^{(2)}$ (Distinct recorded instances of an element within a space).
Powers	$(s_F)^p$	$(s_0^{(1)})^2$ or $(s_\rho)^2$ (Naked superscript applied to the parenthetical element).
Absolute Projection	$\pi_{\infty \rightarrow [\underline{\phi}; \overline{\phi}]}$	Projection from true state to a finite filter.
Relative Projection	$\pi_{[\underline{\phi}_1; \overline{\phi}_1] \rightarrow [\underline{\phi}_2; \overline{\phi}_2]}$	Coarsening from one filter to a narrower filter subset.
Specific/General Quotient Space	$Q_B/Q_{\mathcal{B}}$	Space localised to a specific Borel region B / The family of all quotient spaces across the Borel σ -algebra $\mathcal{B}$.
Specific/General Quotient Map	$\pi_{\mathbb{S} \rightarrow Q, B}, \pi_{\mathbb{S} \rightarrow Q, \mathcal{B}}$	Localisation operator mapped to specific region B / The class of all spatial localisation operators.

This notation presumes an unbounded state-component count for describing a physical scene. While an unbounded count is unlikely in practice (especially at the effective-theory scale), the notation accommodates components not included in this work. A possible extension is that base components depend on resolution, in which case $\mathbb{S}_{n \in N[\epsilon]}$ is an appropriate convention.

Table 1: Notation Summary

Feature	Method 1: Direct Quotient via Local Section Maps	Method 2: Configuration-Space via Global Equivalence Classes
Initialisation	Instantiate an identical space filtered for fields of interest (section mappings).	Map all field variations to a global configuration space (each point an entire universal state).
Grouping Logic	Evaluate field variations directly inside the Borel set B .	Group global points to an equivalence class ($\sim_B$) if Borel coordinates agree.
Quotienting	Map local variations to a single point if the Borels agrees, collapsing fields directly.	Map the entire global equivalence class of configurations to a single point in the quotient space.
Global Space Interaction	None. Skips global universe points entirely.	Direct. The configuration space houses the equivalence class as a region.
Mathematical Equivalence	Evaluated locally over the filtered base space.	$s_1 \sim_B s_2 \iff s_1(\mathbf{x}) = s_2(\mathbf{x}) \quad \forall \mathbf{x} \in B$
Tractability	Possible to obtain finite memory bounds and to run standard tensor operations/PDEs.	Seemingly intractable. Requires simulating variations of the entire physical universe outside Borel sets.
Optimal Use Case	Algorithmic.	Theoretical documentation and topological proofs.

Table 2: Comparison of quotienting methods for isolating local physical structures within Borel spatial bounds.